# Task 4 – Student Dashboard & KPI Measurement

## 1. Objective

The goal is to design a **centralized student dashboard** that consolidates all daily, weekly, monthly, and quarterly activities of Alcovians. This dashboard should:

- Track **quantity** (Did Alcovia conduct the activity?).
- Measure **quality** (Did Alcovians like it?).
- Assess **performance** (How did students perform?).
- Provide **decision-making insights** for stakeholders (parents, advisors, management).
- Minimize manual intervention by automating data capture, storage, and refresh cycles.

## 2. Data Architecture

- **Primary Storage:** All data will be populated into **SQL Server** (or whichever database the application currently using). This ensures structured storage, relational integrity, and scalability.
- **Data Sources:**
    - **App usage logs** (Focus Timer minutes, Realm activity, syllabus completion).
    - **ClickUp inputs** (tasks, goals, advisor notes).
    - **Workshop & Pod session attendance** (via Razorpay + manual logs).
    - **Wearables integration** (Apple Watch, Samsung, Garmin, WHOOP).
    - **Campaign funnel data** (Meta ads, discovery calls, interviews).
- **Data Flow:**
    - Raw data ingested into SQL tables.
    - ETL pipelines (via SSIS or ADF) clean and transform inputs.
    - Data stored in **fact tables** (events, sessions, workshops) and **dimension tables** (students, advisors, activities).

## 3. Visualization Layer

- **Tool:** Power BI will be the visualization and reporting layer.
- **Dashboards:**
    - **Home Screen:** Category breakdown vs Top 30 metrics (e.g., most engaged students, most popular features).
    - **Study Screen:** Syllabus completion %, chapter readiness scores, exam countdowns.
    - **Tasks Screen:** Task completion rate, deadline adherence, time estimation accuracy.
    - **Roadmap Screen:** Academic, personal, and career roadmap progress.
    - **Realm Screen:** Focus Timer minutes, streaks, rankings, dragon collection.
    - **Notifications:** Engagement with nudges, CTR of reminders, drop-off analysis.
- **Decision-Making Visuals:**
    - KPI cards (Did Alcovia conduct it? Did students like it? How did they perform?).
    - Trend charts (weekly focus minutes, syllabus completion trajectory).
    - Funnel visualizations (lead → qualified → discovery call → conversion).
    - Workshop impact analysis (attendance vs feedback vs performance).

## 4. Automation & Data Refresh

- **Short Term:**
    - Manual uploads of CSVs from ClickUp, Razorpay, and app logs into SQL Server.
    - Scheduled refresh in Power BI (daily/weekly).
- **Long Term:**
    - **Automated pipelines** using N8N or ADF to push data directly into SQL Server.
    - Power BI datasets configured with **incremental refresh**.
    - Stakeholders can build **self-service dashboards** on top of curated datasets.

This ensures minimal human intervention and guarantees that insights are always up-to-date.

## 5. Stakeholder Access

- **Students:**
    - Personalized dashboards showing focus minutes, syllabus completion, streaks, and rankings.
    - Motivational nudges based on data (e.g., “You’re 80% ready for your next exam!”).

- **Parents:**
  - Progress reports on academic roadmap, extracurricular engagement, and wellness metrics.
  - Alerts for missed deadlines or declining performance.
- **Advisors:**
  - Consolidated view of all students.
  - Tools to identify at-risk students and recommend interventions.
- **Management:**
  - GTM tracker (revenue, funnel, campaign performance).
  - RCA dashboards to decide which campaigns to scale, refresh, or drop.

### 6. Key Principles

- **Single Source of Truth:** SQL Server (or current DB) is the master repository.
- **Automation First:** Minimize manual uploads, maximize scheduled refresh.
- **Scalability:** Dashboards designed to handle growing student base.
- **Actionable Insights:** Every visual tied to a decision (student motivation, advisor intervention, campaign scaling).
- **Self-Service:** Stakeholders empowered to build their own dashboards without developer dependency.